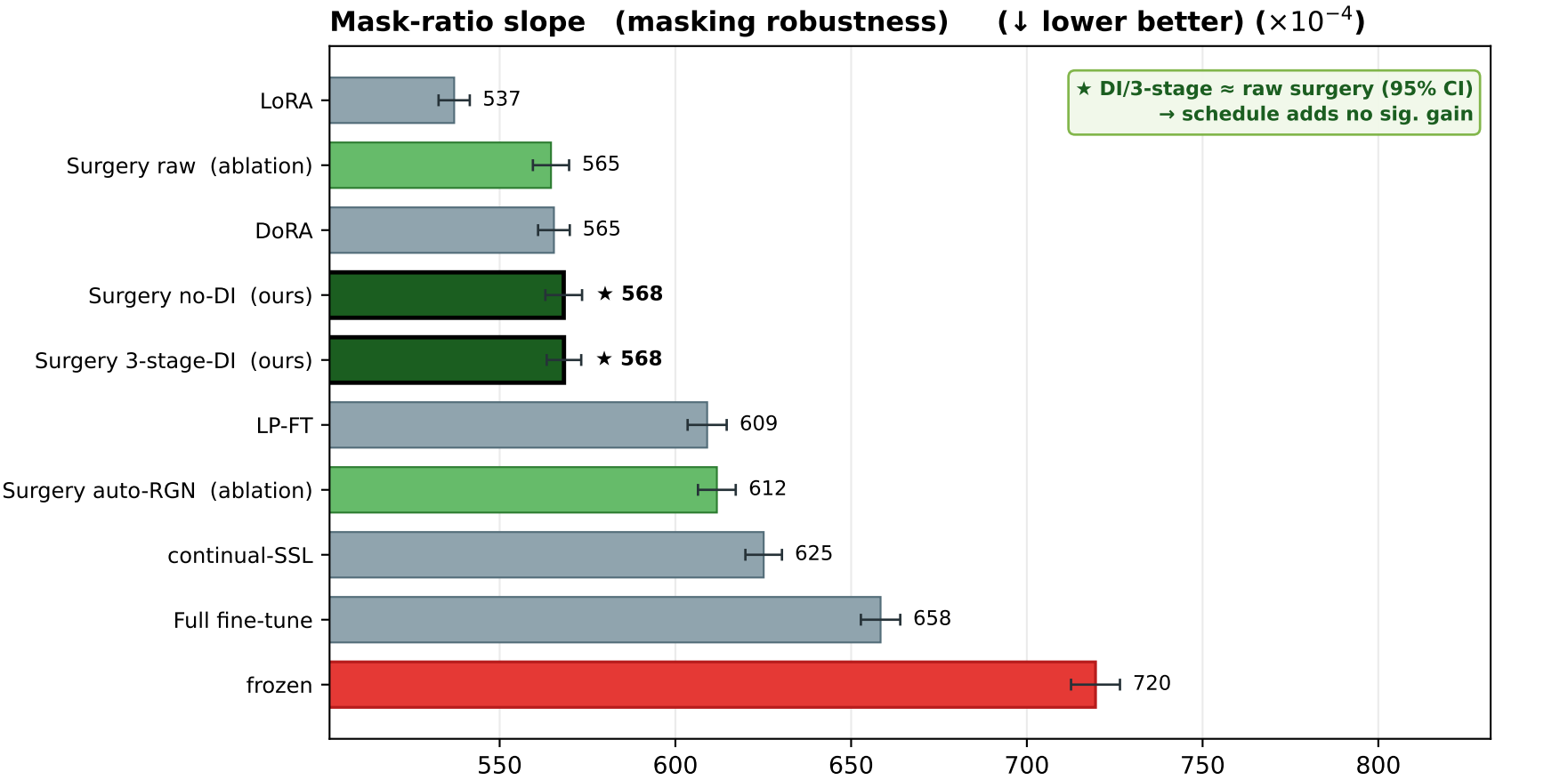
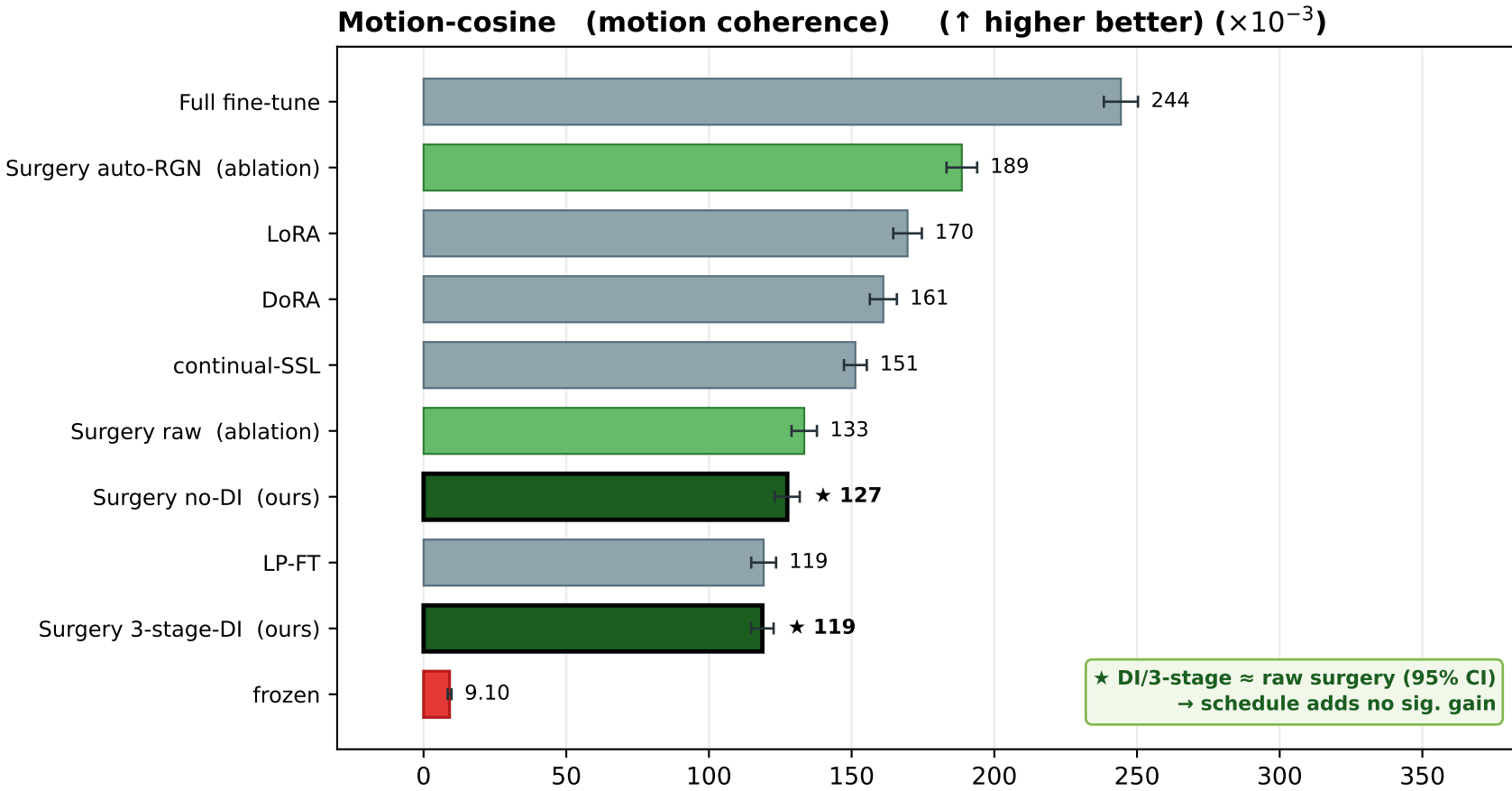
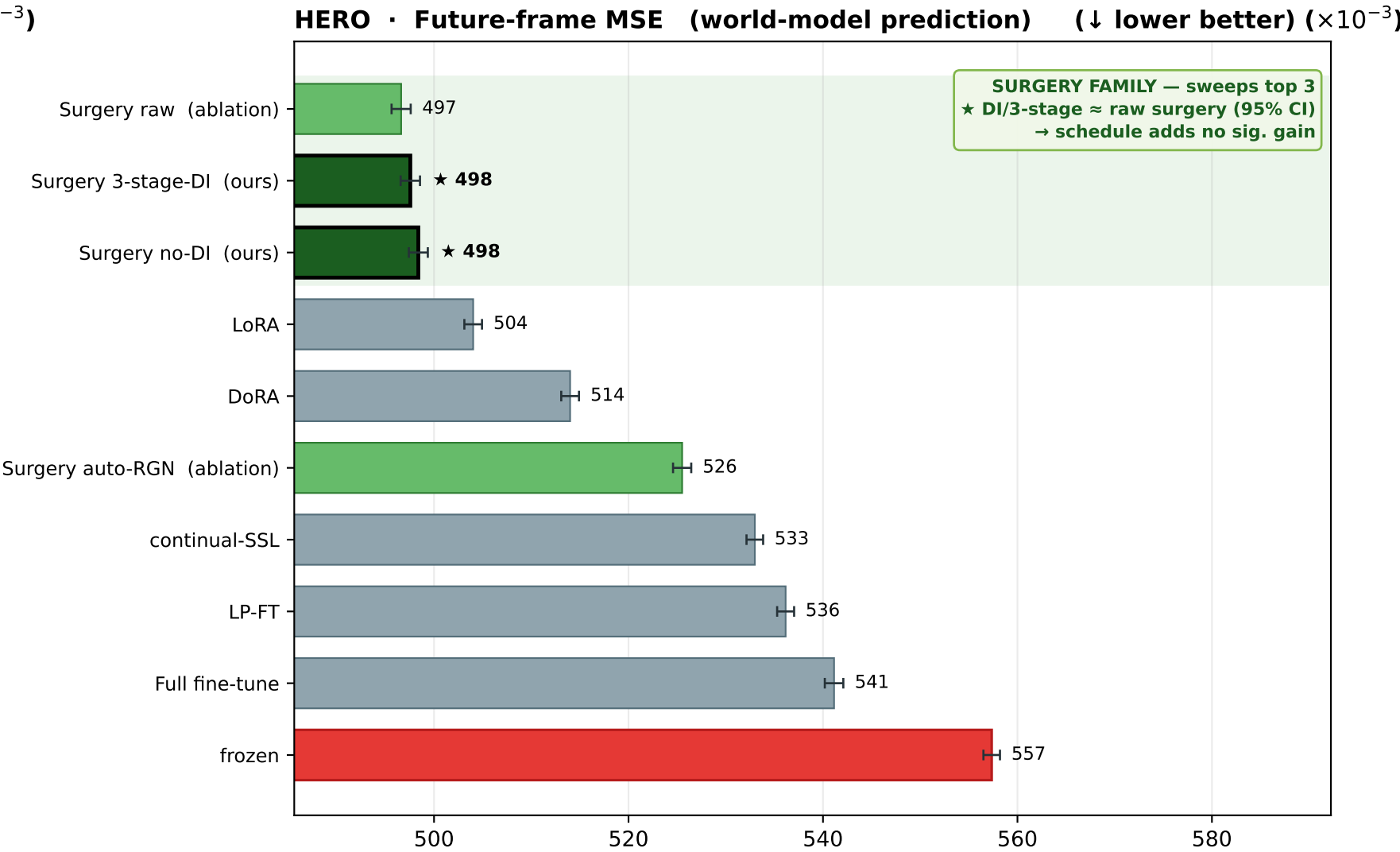
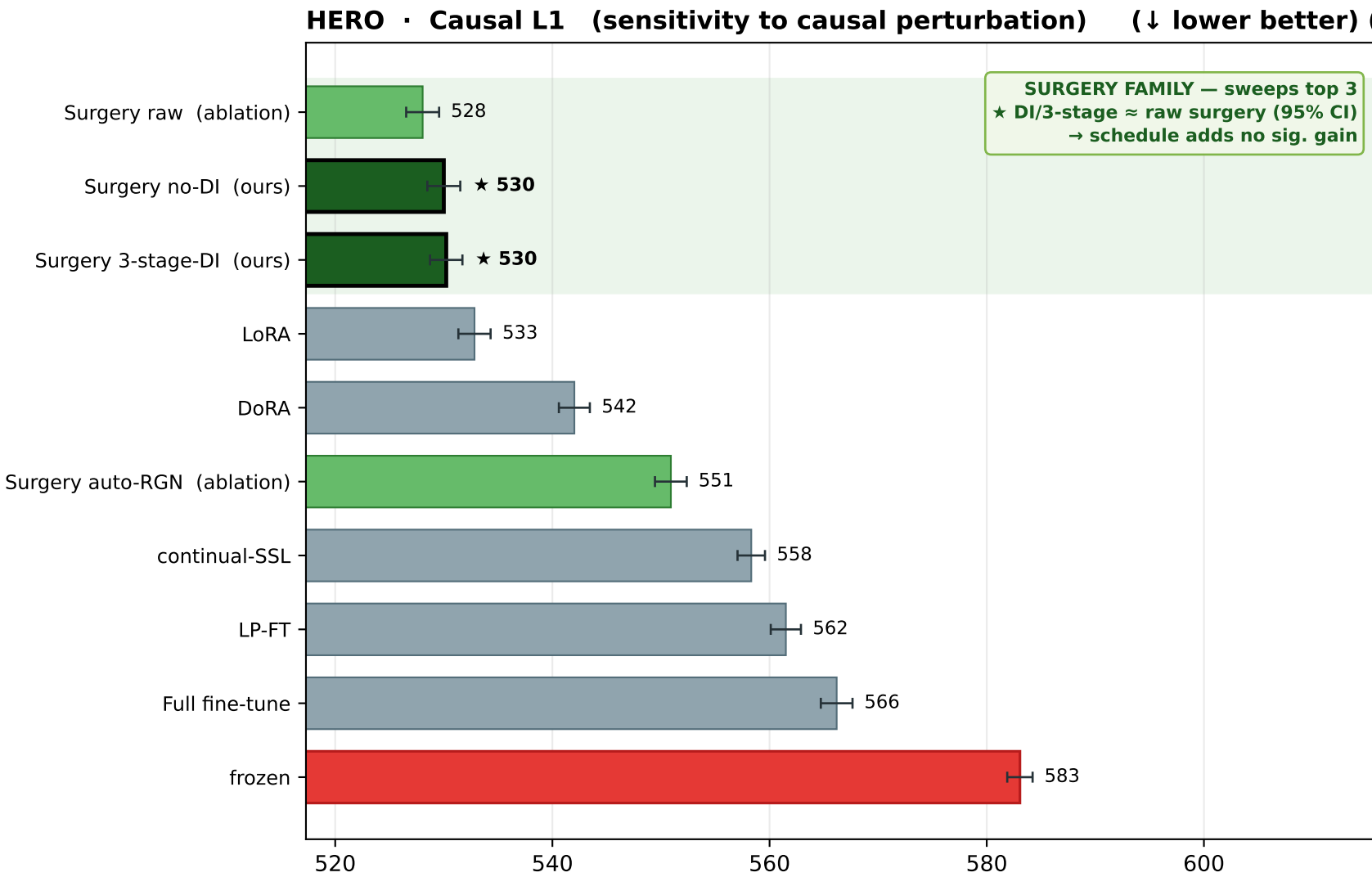


Factor surgery dominates predictive-quality metrics — full baseline set, no per-panel hiding (★ = ours · green band = surgery family · 95% BCa CI)



■ Surgery — ours (3-stage-DI, no-DI) ■ Surgery — ablations (raw, auto-RGN) ■ Fine-tune / PEFT / continual-SSL baselines ■ Frozen V-JEPA (the baseline to beat)

HONEST READ (full baseline set — nothing hidden): the FACTOR-SURGERY FAMILY occupies the top of every predictive-quality metric — causal sensitivity and future-frame MSE — with 95% CIs clear of full fine-tune, LoRA/DoRA and the frozen encoder.

The DI / 3-stage schedule (ours) MATCHES raw surgery within 95% CI: an honest ablation, not a hidden weakness — the gain comes from surgery itself. Surgery trades a small temporal-ordering / TCC margin (see full 14-panel scorecard) for this predictive lead; we report that trade-off rather than dropping those panels.